

Long-term Results of Proton Beam Irradiated Uveal Melanomas

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Abstract: The first 128 consecutive patients with uveal melanomas treated with proton beam irradiation were studied in order to evaluate survival and visual acuity status of patients with relatively long-term follow-up. The median follow-up was 5.4 years, and no patient was lost to follow-up. All tumors showed regression. The most recent visual acuity was 20/40 or better in 35% and 20/100 or better in 58%. Eight eyes were enucleated because of complications. Metastasis developed in 26 patients (20.5%) from 3 months to 7 years after treatment. Results indicate that proton irradiation is quite successful for achieving local control of uveal melanomas. A large proportion of the treated eyes maintained useful vision. Five-year follow-up data indicate that proton irradiation has no deleterious effect on the likelihood of the development of metastasis. [Key words: proton beam irradiation, uveal melanoma.] Ophthalmology 94:349-353, 1987

We have used proton beam irradiation for the treatment of uveal melanomas at the Harvard Cyclotron since 1975, and our results over the first 7½ years were reported recently.¹ Because complications from irradiation as well as metastasis from melanomas may occur years after treatment, we evaluated the results on the first 128 consecutive patients treated in order to assess the effect of relatively long-term follow-up on survival and visual outcome.

PATIENTS AND METHODS

All patients who were treated between July 1975 and December 1981 were identified, and follow-up data were analyzed up to July 31, 1986. No patients were excluded from the analysis.

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CLINICAL DATA

The diagnosis of uveal melanoma was established in all cases through (1) clinical examination by multiple experienced consultants using indirect ophthalmoscopy and slit-lamp biomicroscopy with contact lens and (2) assessment of the results of ancillary tests such as standard and wide-angle fundus photography, fluorescein angiography, and ultrasonography (both immersion and contact A- and B-scan). Radioactive phosphorus (P32) uptake measurements were done routinely only in the initial stages of the study. Documentation of growth was required for small lesions in all but three cases, which were treated early in this series. All the patients received a complete systemic examination by an internist to exclude metastasis or other primary malignancy. Chest roentgenography and liver function studies were performed routinely; a liver scan was performed in all patients with abnormal liver function studies or palpable liver. Patients with uveal melanomas up to 24 mm in largest diameter and 14 mm in height were treated. Largest diameters up to the axial length of the eye were observed in ciliochoroidal melanomas extending circumferentially.

The technical aspects of the operation for the localization of melanomas, the treatment planning, and the tech-